

CSES – Advanced Techniques Problem Reference

All tasks: 1 s, 512 MB. Counts are modulo $10^9 + 7$ only where explicitly stated.

1628 – Meet in the Middle

Given n positive integers, count subsets of the array elements whose sum is exactly x ; equal values at different positions are distinct choices. **In:** n x ; $t_1 \dots t_n$

Out: number of subsets with sum x

Constraints: $n \leq 40$; $x, t_i \leq 10^9$. Use 64-bit count.

2136 – Hamming Distance

Given n binary strings of equal length k , find the minimum Hamming distance between any two distinct strings. **In:** n k ; then n bit strings

Out: minimum number of differing positions

Constraints: $n \leq 2 \cdot 10^4$; $k \leq 30$.

3360 – Corner Subgrid Check

An $n \times n$ grid uses the first k uppercase letters. For each letter, determine whether some subgrid of height and width at least 2 has that letter in all four corners. **In:** n k ; then n grid rows

Out: k lines: YES/NO for each letter

Constraints: $n \leq 3000$; $k \leq 26$.

2137 – Corner Subgrid Count

In an $n \times n$ binary grid, count subgrids of height and width at least 2 whose four corner cells are all black (1). **In:** n ; then n binary rows

Out: number of beautiful subgrids

Constraints: $n \leq 3000$. Use 64-bit count.

2138 – Reachable Nodes

Given a directed acyclic graph, for every node count how many nodes are reachable from it, including the node itself.

In: n m ; then m directed edges a b

Out: n reachability counts

Constraints: $n \leq 5 \cdot 10^4$; $m \leq 10^5$.

2143 – Reachability Queries

Given a directed graph, answer whether node b is reachable from node a for each query. **In:** n m q ; m directed edges; then q pairs a b

Out: YES or NO per query

Constraints: $n \leq 5 \cdot 10^4$; $m, q \leq 10^5$.

2072 – Cut and Paste

Maintain a string under operations that cut the current substring at positions $[a, b]$, remove it, and append it to the end. Output the final string. **In:** n m ; uppercase string; then m lines a b

Out: final string

Constraints: $n, m \leq 2 \cdot 10^5$.

2073 – Substring Reversals

Maintain a string under operations that reverse the current substring at positions $[a, b]$. Output the final string. **In:** n m ; uppercase string; then m lines a b

Out: final string

Constraints: $n, m \leq 2 \cdot 10^5$.

2074 – Reversals and Sums

Maintain an integer array with operations: 1 a b reverses subarray $[a, b]$; 2 a b asks its sum. **In:** n m ; array; then m operations

Out: sum for every type-2 operation

Constraints: $n \leq 2 \cdot 10^5$; $m \leq 10^5$; $0 \leq x_i \leq 10^9$. Use 64-bit sums.

2076 – Necessary Roads

A connected simple undirected graph is given. Find every road whose removal makes some pair of cities disconnected.

In: n m ; then m undirected edges

Out: number of necessary roads, then the roads in any order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2077 – Necessary Cities

A connected simple undirected graph is given. Find every city whose removal, together with its incident roads, leaves some two other cities disconnected. **In:** n m ; then m undirected edges

Out: number of necessary cities, then their list in any order

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2078 – Eulerian Subgraphs

Choose any subset of edges of an undirected graph while keeping all vertices. Count choices for which every vertex has even degree. **In:** n m ; then m undirected edges

Out: number of Eulerian subgraphs modulo $10^9 + 7$

Constraints: $n \leq 10^5$; $m \leq 2 \cdot 10^5$.

2084 – Monster Game I

Process levels in order. At level $i < n$, kill or escape; level n must be killed. Killing costs $s_i \cdot f$ using the current skill factor f , then sets $f = f_i$; escaping changes nothing. Minimize total time. **In:** n x ; strengths s_i ; resulting skill factors f_i

Out: minimum total time

Constraints: $n \leq 2 \cdot 10^5$; $s_i \leq 10^6$ nondecreasing; $x \geq f_1 \geq \dots \geq f_n \geq 1$. Use 64-bit.

CSES – Advanced Techniques Problem Reference

Continued. All tasks: 1 s, 512 MB.

2085 – Monster Game II

Same game as Monster Game I: kill/escape levels in order, final monster mandatory; killing i costs s_i times the current skill factor and then sets it to f_i . Here strengths and skill factors have no monotonicity guarantees. **In:** $\mathbf{n\ x}$; strengths s_i ; resulting factors f_i

Out: minimum total time

Constraints: $n \leq 2 \cdot 10^5$; $x, s_i, f_i \leq 10^6$. Use 64-bit.

2086 – Subarray Squares

Partition an array into exactly k nonempty contiguous subarrays. A part costs the square of the sum of its elements; minimize the total cost. **In:** $\mathbf{n\ k; x_1 \dots x_n}$

Out: minimum total cost

Constraints: $1 \leq k \leq n \leq 3000$; $x_i \leq 10^5$. Use 64-bit.

2087 – Houses and Schools

Houses are at positions $1 \dots n$ with c_i children at house i . Place k schools in houses; every child walks to a nearest school. Minimize total walking distance. **In:** $\mathbf{n\ k; c_1 \dots c_n}$

Out: minimum total distance

Constraints: $1 \leq k \leq n \leq 3000$; $c_i \leq 10^9$. Use 64-bit.

2088 – Knuth Division

Starting with the whole array as one subarray, repeatedly split one current subarray into two until all parts are singletons. Each split costs the sum of the values in the subarray being split. Minimize total cost. **In:** $\mathbf{n; x_1 \dots x_n}$

Out: minimum total split cost

Constraints: $n \leq 5000$; $x_i \leq 10^9$. Use 64-bit.

2111 – Apples and Bananas

There are n apples and m bananas, each with integer weight $1 \dots k$. For every total weight $w = 2 \dots 2k$, count pairs consisting of one apple and one banana with combined weight w . **In:** $\mathbf{k\ n\ m}$; apple weights; banana weights

Out: $2k - 1$ counts for totals $2 \dots 2k$

Constraints: $k, n, m \leq 2 \cdot 10^5$. Use 64-bit counts.

2112 – One Bit Positions

Given a binary string, for every distance $d = 1 \dots n - 1$, count pairs of positions i, j with $i - j = d$ and both characters equal to 1. **In:** one binary string of length n

Out: $n - 1$ counts, for distances $1 \dots n - 1$

Constraints: $2 \leq n \leq 2 \cdot 10^5$.

2113 – Signal Processing

Slide a mask across a signal from left to right, including partial overlaps at both ends. At every alignment, output the sum of products of all aligned signal/mask values. **In:** $\mathbf{n\ m}$; signal $a_1 \dots a_n$; mask $b_1 \dots b_m$

Out: $n + m - 1$ overlap dot products in order

Constraints: $n, m \leq 2 \cdot 10^5$; $a_i, b_i \leq 100$. Use 64-bit sums.

2101 – New Roads Queries

Initially there are no roads; one undirected road is added each day in the given order. For each pair (a, b) , find the first day when a and b become connected. **In:** $\mathbf{n\ m\ q}$; m roads in build order; then q pairs

Out: earliest day for each query, or -1 if never connected

Constraints: $n, m, q \leq 2 \cdot 10^5$.

2133 – Dynamic Connectivity

Maintain an undirected simple graph under edge insertions and deletions. Report the number of connected components before any event and after every event. **In:** $\mathbf{n\ m\ k}$; m initial edges; then k events $\mathbf{t\ a\ b}$

Out: $k + 1$ component counts

Constraints: $n \leq 10^5$; $m, k \leq 10^5$; updates are always valid.

2121 – Parcel Delivery

A directed route $a \rightarrow b$ has capacity r parcels and cost c per parcel. Send exactly k parcels from city 1 to city n at minimum total cost. **In:** $\mathbf{n\ m\ k}$; then routes $\mathbf{a\ b\ r\ c}$

Out: minimum cost, or -1 if sending k parcels is impossible

Constraints: $n \leq 500$; $m \leq 1000$; $k \leq 100$; $r, c \leq 1000$.

2129 – Task Assignment

There are n employees and n tasks with cost c_{ij} for assigning task j to employee i . Assign every employee exactly one distinct task, minimizing total cost, and output one optimal assignment. **In:** $\mathbf{n}$; then $n \times n$ cost matrix

Out: minimum cost; then n employee–task pairs

Constraints: $n \leq 200$; $1 \leq c_{ij} \leq 1000$.

2130 – Distinct Routes II

A directed graph has teleporters that may each be used at most once over the entire game. Play exactly k days; each day starts at node 1 and must reach node n . Every teleporter use costs one coin. Minimize total coins and output the k routes. **In:** $\mathbf{n\ m\ k}$; then m directed teleporters $\mathbf{a\ b}$

Out: minimum total uses and k valid routes, or only -1 if impossible

Constraints: $n \leq 500$; $m \leq 1000$; $1 \leq k \leq n - 1$.